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Oei Tiong Ham Building
Singapore 259772
Tel: (65) 6516 6134 Fax: (65) 6778 1020
Website: www.lkyspp.nus.edu.sg

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Policy learning, evaluation, and aid effectiveness: Mining lessons on project and program success from the Asian Development Bank

Michael Howlett
Lee Kuan Yew School of Public Policy, National University of Singapore &
Department of Political Science, Simon Fraser University
Email: sppmph@nus.edu.sg

Nihit Goyal
Lee Kuan Yew School of Public Policy
National University of Singapore
Email: nihit@u.nus.edu

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Lee Kuan Yew School of Public Policy
National University of Singapore

469C Bukit Timah Road
Oei Tiong Ham Building
Singapore 259772
Tel: (65) 6516 6134 Fax: (65) 6778 1020
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Abstract

Lesson drawing, or learning from past policies or programs, can improve current or future policies or programs and, thereby, lead to policy success. This requires various types of evaluations that identify and highlight different causal relationships in the system. However, the literature on policy evaluation has little to say about how such evaluations work in practice. For example, in the case of overseas development assistance, although multiple studies examine factors that contribute to aid effectiveness, they do not use and build on lessons from internal evaluations of aid projects and programs. Using data on project and program evaluations from the Asian Development Bank (ADB), this paper compares the lessons from external evaluations on aid effectiveness with those of internal evaluations. It critically examines 'lessons learned' by the ADB in over 950 sovereign interventions across 38 countries in Asia-Pacific during 1996-2016 using relatively new 'data science' approach of text mining. It specifically analyzes term frequencies, proportions in evaluations of successful and unsuccessful interventions, and correlations to understand the content and content relationships of the lessons learnt. It finds that while internal evaluations validate and even go beyond several micro and meso level lessons of external evaluations – such as within country and sector variation and project characteristics of (un)successful interventions – they say less about macro level, theoretical relationships that set the context for aid effectiveness, such as per capita economic growth or the level of democracy in the borrowing country. The findings suggest the need for a multilevel evaluation framework consisting of micro, meso, and macro evaluations which pick up different factors that influence success and failure and, hence, contribute to better lesson drawing.

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1. Introduction

The literature on policy learning has emphasized the role of learning in the policy process and policy change. Hall (1993), for example, argued that typical feedback and learning from policy or program execution often resulted in incremental, first- or second-order policy change while social learning to address policy anomalies resulted in paradigmatic, third-order change. Similarly, Rose (1991) posited that lessons can be drawn from both past policies in the same jurisdiction as well as from ongoing policies in other jurisdictions to address social problems and reduce dissatisfaction.

Appropriate evaluation is important for such learning to occur (Sanderson, 2002). This is true not only for the policy as a whole, but also for the programs and projects that implement the policy (McConnell, 2010). This is particularly significant in case of overseas development assistance, which amounts to over USD 150 billion annually. Multilateral development agencies conduct various types of evaluations to analyze the effectiveness of foreign aid. The Asian Development Bank (ADB), for example, has numerous types of project or program evaluations – project completion reports, validation of project completion reports, project or program evaluation reports, technical assistance completion reports, and technical assistance project evaluation reports.

External studies such as those by Bulman, Kolkma, and Kraay (2017) and Feeny and Vuong (2017) examine factors that influence project and program success through econometric analyses of ADB and World Bank project evaluation database. However, research has scarcely examined the "lessons learned" by the project team or the evaluation department, usually a

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component of self- or independent evaluation, to better project or program success and aid effectiveness.

In fact, the ADB Annual Evaluation Review (AER) 2017 noted the importance of learning from past lessons:

> Value is also to be derived from transferring experiences from one project to another… Effective learning from the depth and breadth of operational experience is a valuable source of comparative advantage for ADB, which the institution can build on as it matures as a knowledge bank. In 2007, an IED study showed that the characteristics of successful projects included the ability to learn from past lessons and incorporate them in project design, and that quality at entry depends on the incorporation of lessons from past projects.

Despite this, ADB AER 2017 found that even within the ADB documented lessons were under-utilized for learning: "as an illustration, 20% of staff knew of projects financed through ADB that failed to deliver on intended results because lessons were ignored."

While it was previously cumbersome to examine the content of a large number of documents containing unstructured text previously, advances in text analysis make such examination possible. This exploratory study investigates how lessons learnt in internal evaluations of the ADB compare with findings of external studies on aid effectiveness using text mining.

Besides contributing to the literature on aid effectiveness, this study contributes to the research on policy learning and evaluation by suggesting a role for a multilevel evaluation framework for projects, programs, and policies for greater program and policy success. In addition, it also

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contributes to the literature on methods of comparative policy analysis by testing the use of a method for analyzing a large volume of unstructured text data.

2. What others have learnt: Lessons from external evaluations of aid effectiveness

Existing literature on aid effectiveness has highlighted numerous characteristics that influence project or program success. In an early study on the topic, Deininger, Squire, and Basu (1998) found that country- and sector-specific research was a predictor of the quality of World Bank foreign assistance. They argued that a transfer of resources from project preparation and supervision to research on the project context would improve aid effectiveness. This implied that the likelihood of project and program success would be higher when the World Bank had a better understanding of the country and the sector of operation. Feeny and Vuong (2017), too, have found an association between the recipient country and sector of operation and project or program success in case of the ADB, while Mubila, Lufumpa, and Kayizzi-Mugerwa (2000) have corroborated the correlation between sector of operation and aid effectiveness using data from the African Development Bank.

Hypothesis H1: Project or program success is associated with country and sector of operation.

Scholars have found that higher per capita Gross Domestic Product (GDP) or its growth rate is associated with higher project or program success across the African Development Bank, the ADB, and the World Bank (Bulman et al., 2017; Denizer, Kaufmann, & Kraay, 2013; Feeny & Vuong, 2017; Mubila et al., 2000). Lower inflation has also been associated with higher

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likelihood of project or program success (Mubila et al., 2000). This may also be due to lack of adequate resources to complement foreign aid or the presence of macroeconomic policies that are correlated with GDP growth rate as well as rate of return on foreign aid (Isham & Kaufmann, 1999).

Hypothesis H2: Project or program success is associated with high (per capita) GDP, or its growth rate.

Dollar and Levin (2005), however, have suggested that per capita GDP growth rate may itself be endogenous, reflecting a better institutional and policy environment in the recipient country. Scholars have examined the effect of different measures of institutional and policy environment on project and program success. As mentioned earlier, higher economic rate of return on multilateral investment may reflect better macroeconomic management and policymaking in the borrowing country, in the form of 'undistorted' macroeconomic, exchange rate, trade, and pricing policies (Isham & Kaufmann, 1999). Similarly, well-defined property rights and the rule of law may influence both project or program success and real per capita GDP growth rate (Dollar & Levin, 2005). Recent studies have used more aggregate indicators such as the Country Policy and Institutional Assessment (Bulman et al., 2017; Denizer et al., 2013; Geli, Kraay, & Nobakht, 2014) to explain aid effectiveness. The Country Policy and Institutional Assessment rates countries based on economic management, structural policies, policies for social inclusion, and public sector management. Feeny and Vuong (2017) have found that while the World Governance Indicators

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index, as a whole, is not correlated with project or program outcome after controlling for micro level characteristics, the subindex on political stability is positively related to aid effectiveness.

Hypothesis H3: Project or program success is associated with better policy and institutional environment.

Although scholars have also posited that political rights and civil liberties influence project or program success, the findings have been mixed. Isham, Kaufmann, and Pritchett (1997) found that stronger civil liberties were associated with higher economic rate of return in case of government investment, more generally, and posited that this may be attributable to the beneficial effect of citizen voice and public accountability on project outcomes. In his study on the impact of aid on quality of life, Kosack (2003) found that the political environment in a country influenced aid outcomes – while in democracies aid improved quality of life, in autocracies it was ineffective at best or harmful at worst. Subsequently, Dollar and Levin (2005) found that democracies had a better record of success for projects or programs supported by the World Bank. Recent studies on aid effectiveness in Asia have found that civil liberties and political freedom are negatively correlated with project or program success (Bulman et al., 2017; Feeny & Vuong, 2017) and suggested that the strong performance of China and Viet Nam may account for this anomaly. However, even Denizer et al. (2013) found that the relationship between aid effectiveness and civil liberties does not hold after controlling for project characteristics. Another plausible explanation for this contradiction was provided by Wane (2004), who posited that governments with high accountability and capacity were less likely to select projects with appropriate design, while

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governments with low accountability or capacity were more likely to accept poorly designed projects. This is consistent with Cruz and Keefer (2015), who have argued that clientelist politics and non-programmatic parties, and not the electoral system, are a better predictor of success in bureaucratic reform as clientelist politicians oppose aid that reduces their patronage power and have weaker incentive to hold the executive branch accountable.

Hypothesis H4a: Project or program success in Asia is associated with weak civil liberties and political freedom.

Hypothesis H4b: Project or program success is associated with adequate political or public accountability.

While the above hypotheses point to more country-level characteristics that affect aid effectiveness, studies have found that within country variation in project or program success is higher than between country variation (Bulman et al., 2017; Denizer et al., 2013; Feeny & Vuong, 2017). Dollar and Levin (2005) have found that the effects of rule of law and civil liberties vary across sectors, explaining some of the within country variation in aid effectiveness. Other scholars have posited that 'micro' level variables influence project or outcomes too. For instance, project characteristics such as longer duration, larger size, and more preparation time are associated with greater complexity and, thus, might correlate with lower likelihood of project or program success (Denizer et al., 2013). On the other hand, larger size might result in greater commitment from the project staff, increasing the likelihood of success (Feeny & Vuong, 2017). As the difference between approved funding and actual expenditure is negatively correlated with project or program

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success, Bulman et al. (2017) have also suggested that unsuccessful interventions are likely to be terminated early while Feeny and Vuong (2017) have offered capacity constraint as an explanation.

Hypothesis H5: Project or program success is associated with project or program specific characteristics such as less complexity, greater commitment, lower likelihood of early termination, and fewer capacity constraints.

Finally, project management and supervision has been identified as a key determinant of project or program success. Kilby (2000) found that the benefit of additional project supervision outweighed its costs for World Bank projects or programs. Meanwhile, Chauvet, Collier, and Duponchel (2010) noted that project supervision affected project or program success even in a post-conflict context. Better track record of the project manager, or the Task Team Leader in case of the World Bank, has been found to be positively correlated with project or program success (Bulman et al., 2017; Denizer et al., 2013; Geli et al., 2014). Feeny and Vuong (2017) have posited that weak management capacity may result in lower fund disbursement and lesser likelihood of success.

Hypothesis H6: Project or program success is associated with better project management and supervision.

While H1, H2, and H4a reflect more macro level characteristics of the recipient or borrowing country, H3 and H4 are likely to operate both at the macro and the meso level, and H5 and H6 are

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more pertinent at the project level. Although H1-H4 have been examined in the literature, few studies have tested hypotheses H5-6 directly, often relying on these characteristics as explanatory mechanisms for the constructs tested. However, our dataset allows us to test these directly and also compare their relevance vis-à-vis H1-H4.

3. Methods

As mentioned earlier, this study is based on data from internal evaluations of projects and programs of the ADB. The ADB is a multilateral development financial institution with a vision to promote social and economic development in the Asia Pacific. Established in 1966, the ADB has funded grants, loans, and technical assistance for development in over 40 countries in the region (Asian Development Bank, 2017a). The institution's project or program operations are classified under ten sectors: agriculture, natural resources, and rural development; education; energy; finance; health; industry and trade; information and communication technology; public sector management; transport; and, water and other urban infrastructure services (Asian Development Bank, 2014).

Our dataset consists of 1,991 evaluation reports from 1989 to 2016. The report types in the dataset include: 'Annual Evaluation Review' (AER; n = 4), 'Annual Report on Portfolio Performance' (ARPP; n = 2), 'Country Assistance Performance Evaluation' (CAPE; n = 30), 'Country Partnership Strategy Final Review' (CPSFR; n = 7), 'Impact Evaluation Study' (IES; n = 24), 'Project/Program Completion Reports' (PCR; n = 915), 'Project/Program Performance Evaluation Report' (PPER; n = 353), 'Technical Assistance Completion Report' (TCR; n = 176),

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'Technical Assistance Performance Evaluation' (TAPE; n = 49), 'Thematic Evaluation Study' (TES; n = 16), and 'Validation of Project/Program Completion Report' (PVR; n = 415).

Of these, PCRs, PPERs, PVRs, TACRs, and TAPEs evaluate projects or programs and are directly relevant to this study. A PCR is a self-evaluation of a grant or loan project/program, conducted by the responsible regional department within two years of project or program completion. The project or program can subsequently be subject to an independent evaluation by the Independent Evaluation Department of the ADB, either through a short PVR or a more detailed PPER. Similarly, a TCR is a self-evaluation of technical assistance project/program, conducted by the responsible regional department within one year of project or program completion, while a TAPE is an evaluation of a technical assistance project or program by the Independent Evaluation Department. While some projects or programs have only been evaluated by the regional department others have also been evaluated by the Independent Evaluation Department.

Inclusion of all above-mentioned reports would, therefore, have resulted in double-counting and possibly bias our analysis. Instead, this study focused on PCRs and TCRs, as these were available for a larger number or projects or programs. Further, the lessons in these pertained more to the project or program while those in PPERs, PVRs, and TACEs also reflected on the quality of the self-evaluation. While lessons of the self-evaluation may raise a concern regarding the 'objectivity' of the findings, it should be noted that nearly 90% of projects or programs evaluated by the Independent Evaluation Department received the same rating in independent evaluation as in the self-evaluation. Although the number of PCRs and TCRs in the database totaled 1084 evaluations, 75 of these were found to be identical to an existing evaluation in the dataset. Therefore, 1009 evaluations (840 PCRs and 169 TCRs) were processed further.

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For each evaluation, information on country, sector, project or program rating, and lessons learnt is recorded in the dataset. A project or program is evaluated based on four criteria: relevance, effectiveness, efficiency, and sustainability. A projector program receives a score between zero and three on each criterion and one of the following as its overall rating: ‘Highly successful’, ‘Successful/Generally successful’, ‘Partly/less than successful’, or ‘Unsuccessful’ (Asian Development Bank, 2006). Following (Asian Development Bank, 2017b), this study used a binary rating in which a project is deemed as successful if it is ‘Highly successful’ or ‘Successful/Generally successful’, and unsuccessful otherwise. The country for multiple country interventions was coded as ‘regional’ and the sector for multiple sector interventions was coded as ‘multisector.’ This information was correlated with the ‘lessons learnt’ in internal evaluations.

The lessons documented in the PCRs and TACRs were about 350-400 words on average. Typically, they reflected on the project or program performance in light of various topics such as project cycle stage, project management, financial aspect, policy reform, methodologies or approaches, capacity development, sector related issues, and cross cutting issues. The text of lessons was processed through the following. First, the text was converted to lower case, typographical errors were identified using Hunspell (version 2.9; FSF.hu Foundation, Hungary), a spell checker used by various software including Google Chrome, Mozilla Firefox, and Mac OS (Hunspell, 2018), and corrected manually. Second, lines with empty lesson topics were removed to prevent topic phrases (such as ‘project cycle stage’ or ‘cross cutting issues’) from being counted when a topic was not identified. Third, whitespace, punctuation, numerals, and non-alphanumeric characters were removed from the text. Fourth, phrases of up to 7 words in length were identified through tokenization and manual parsing and replaced with a single term (such as

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'project_cycle_stage' or 'cross_cutting_issue'). Fifth, different forms of a term (such as singular and plural, or noun and verb) were harmonized manually to reduce complexity while retaining the themes discussed. Sixth, terms that are used frequently in the English language (such as 'and' and 'the'), terms that were not interesting or not insightful (such as 'subproject' or 'subprogram'), or terms that appeared in fewer than five evaluations were removed from the text to allow for more robust and generalizable findings. Finally, evaluations that contained no text were removed from the dataset.

The final sample contained 982 evaluations (813 PCRs and 169 TACRs) spanning 38 countries during 1996-2016. The countries with the largest number of interventions in the sample were: China (n = 108), Indonesia (n = 91), Pakistan (n = 73), Bangladesh (n = 57), Philippines (n = 56), Viet Nam (n = 53), and India (n = 50). Ninety-nine interventions involved multiple countries. Meanwhile, the sectors with the largest number of interventions were: transport (n = 142), agriculture (n = 128), and public sector management (n = 86). While 119 interventions involved multiple sectors, 138 interventions did not contain sector information. The data contained significant variation in project and program success both across and within countries (Figure 1).

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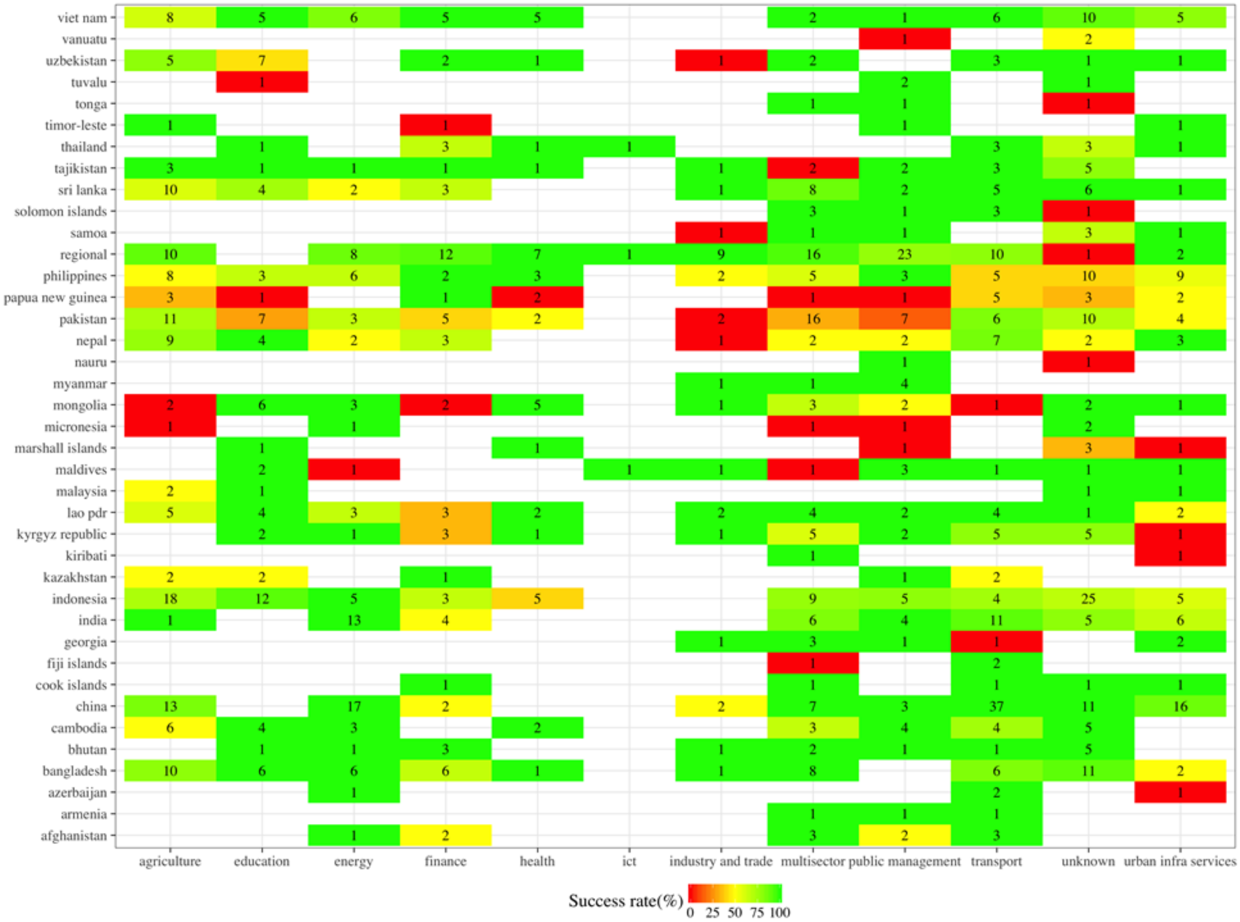

Figure 1: Heat map of project success across countries and sectors. The labels indicate the number of completed projects in that sector in that country. Multicounty projects have been classified as regional. Sector information was not available for 138 projects. The sector for these has been coded as unknown. The color of each cell indicates the success rate for that sector in that country, according to the legend.

This study used text data mining techniques on this ‘corpus’ to test the hypotheses set forth. Text mining is “a knowledge-intensive process in which a user interacts with a document collection… to extract useful information from data sources through the identification and exploration of interesting patterns… the data sources are document collections, and interesting patterns are found not among formalized database records but in the unstructured textual data in

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the documents in these collections" (Feldman & Sanger, 2006, p. 1). While text mining can be conducted for various features – such as characters, words, or terms – this study focused on terms. In particular, it analyzed term frequencies to understand how lesson content varied between evaluations of successful and unsuccessful projects or programs, estimated the log odds of term proportion in evaluations of unsuccessful projects or programs versus evaluations of successful projects or programs[1], and examined co-occurrence metrics between terms to understand content relationships in evaluation lessons. The analysis was conducted using R software (version R 3.4.3; R Foundation for Statistical Computing, Vienna, Austria).

4. What the ADB has learnt: Lessons from internal evaluations

A graph of the term proportions in evaluations of unsuccessful projects or programs versus evaluations of successful projects or programs is shown in Figure 2. This figure and the log odds associated with various country or sector specific terms confirm between country and sector variation in project or program success. For example, countries such as Tajikistan (log odds: -1.64), Bhutan (log odds: -1.57), China (log odds: -1.44), and Viet Nam (-1.40) are mentioned more often in evaluations of successful interventions, while countries such as Pakistan (log odds: 1.93), Marshall Islands (log odds: 1.69), Papua New Guinea (log odds: 1.29) and the Philippines (log odds: 0.79) are mentioned in evaluations of unsuccessful interventions. This is consistent with the high success rates for the first four (85-100%) and the low success rates for the last four (32-59%).

[1] Log odds represents the log of the odds ratio of term proportion in evaluations of unsuccessful projects or programs over term proportion in evaluations of successful projects or programs. A (more) negative value of the log odds indicates that the term has a higher proportion in evaluations of successful interventions than in evaluations of unsuccessful interventions while a (more) positive value indicates the opposite.

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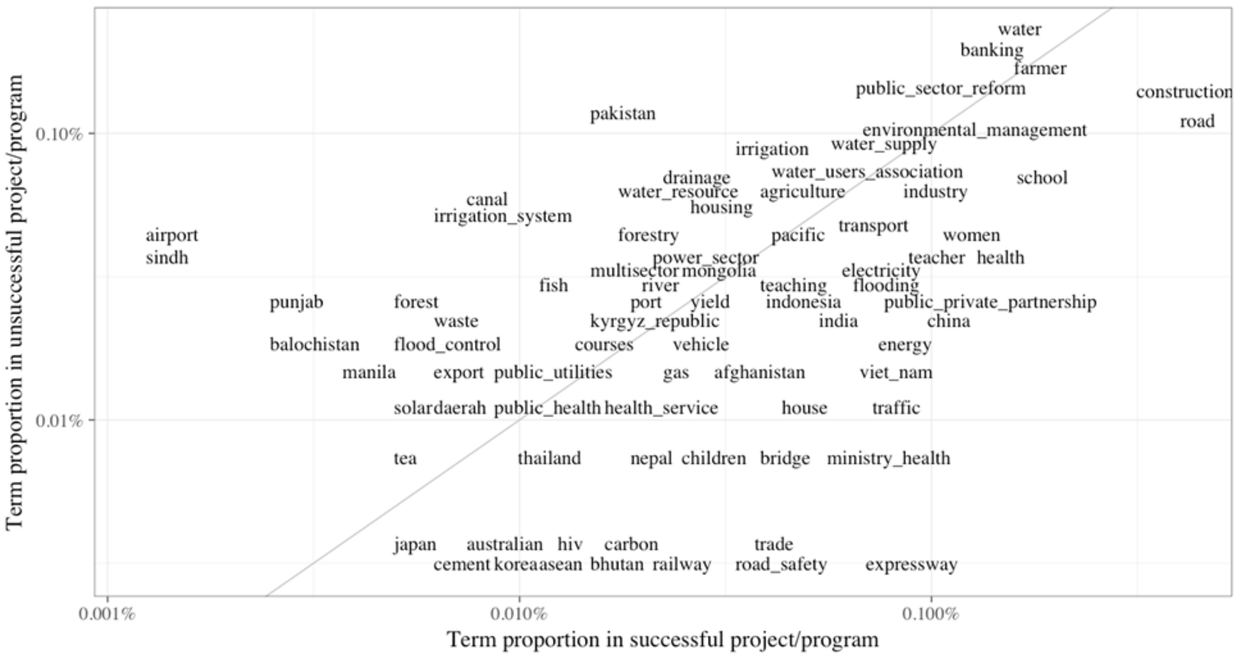

Figure 2: Term proportion in evaluations of unsuccessful interventions versus successful interventions

Similarly, sectors with above average success – energy (89%), transport (87%), and education (80%), for example – received disproportionate mention in evaluations of successful interventions as seen in the use of terms such as 'energy efficiency' (log odds: -1.95), 'energy' (log odds: -1.32), and 'power plant' (log odds: -1.08); 'expressway[s]' (log odds: -3.05), 'road safety' (log odds: -2.33), and 'local road' (log odds: -2.30); and, 'education sector' (log odds: -2.59), 'madrasah' (log odds: -1.78), and 'teacher training' (log odds: -1.64). On the other hand, sectors with less than average success, such as agriculture, natural resources, and rural development (70%) and water and other urban infrastructure services (72%) received mention in both evaluations of successful and unsuccessful interventions. This can be seen in the higher use of terms such as 'farmer group' (log odds: -1.31) 'agricultural development' (log odds: -0.30), and 'wastewater' (log odds: -1.83) but the lower use of terms such as 'agriculture sector' (log odds: 0.71), 'rural development' (log odds: 0.53), and 'water sector' (log odds: 0.59) in evaluations of successful interventions.

However, significantly, while external evaluations typically focus at the country level, internal evaluations suggest that subnational characteristics, for example, due to the government structure or spatial characteristics, influence project and program success as well. Illustratively, terms such as 'municipal government' (log odds: -2.37), 'regional government' (log odds: -2.09), and 'district government' (log odds: -0.10) have been used more often in evaluations of successful interventions, terms such as 'local government unit' (log odds: 2.29), 'provincial government' (log odds: 0.45), 'state government' (log odds: 0.17) have been used more in evaluations of unsuccessful interventions. Similarly, in case of Pakistan, provinces of 'Sindh' (log odds: 2.70), 'Punjab' (log odds: 1.98), and 'Balochistan' (log odds: 1.69), have a higher likelihood of occurrence in unsuccessful interventions. In case of the Philippines, 'Manila' (log odds: 1.22) has a higher likelihood of occurrence in evaluations of unsuccessful interventions.

In addition, while external evaluations focus at the sector level, internal evaluations highlight within sector variation in aid effectiveness as well. For instance, although a high proportion of terms such as 'expressway[s]', 'road safety', (local or rural) 'roads', 'highway' (log odds: -2.26), and 'railway' (-1.89) in evaluations of successful interventions corroborates ADB's track record in land transport, the mention of terms such as 'airport' (log odds: 2.87) and 'port' (log odds: 0.30) in evaluations of unsuccessful interventions suggests that the organization is less successful in air and water transport. Similarly, while the ADB's below par performance in water services is reflected in the disproportionate mention of terms such as 'water resource' (log odds:

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1.18), ‘water supply [and] sanitation’ (log odds: 0.77), ‘water management’ (log odds: 0.71), and ‘water quality’ (log odds: 0.71) in evaluations of unsuccessful interventions, its strength in wastewater treatment is indicated by the higher occurrence of terms such as ‘wastewater’ (log odds: -1.84), ‘wastewater tariff’ (log odds: -1.57), and ‘wastewater treatment plant’ (log odds: -0.55) in evaluations of successful interventions.

Internal evaluations mention terms relevant to economic growth as well. While ‘economic growth’ (log odds: -0.79) and ‘economic development’ (log odds: -0.47) are mentioned more frequently in evaluations of successful interventions, ‘gross domestic product’ (log odds: 0.84) is mentioned more frequently in evaluations of unsuccessful interventions. A closer examination terms highly correlated with ‘economic development’, ‘economic growth’, and ‘gross domestic product’ suggests that economic development is discussed in the context of lending strategy (project-based versus programmatic), economic growth influences fiscal aspects of interventions, such as revenue collection or transaction costs, and GDP is often used in evaluations of interventions associated with macroeconomic policies (Figure 3). Although the term correlation graph for ‘economic growth’ highlights potential causal mechanisms for its observed relationship with project or program success in the literature, internal evaluations do not offer sufficient evidence to support H2.

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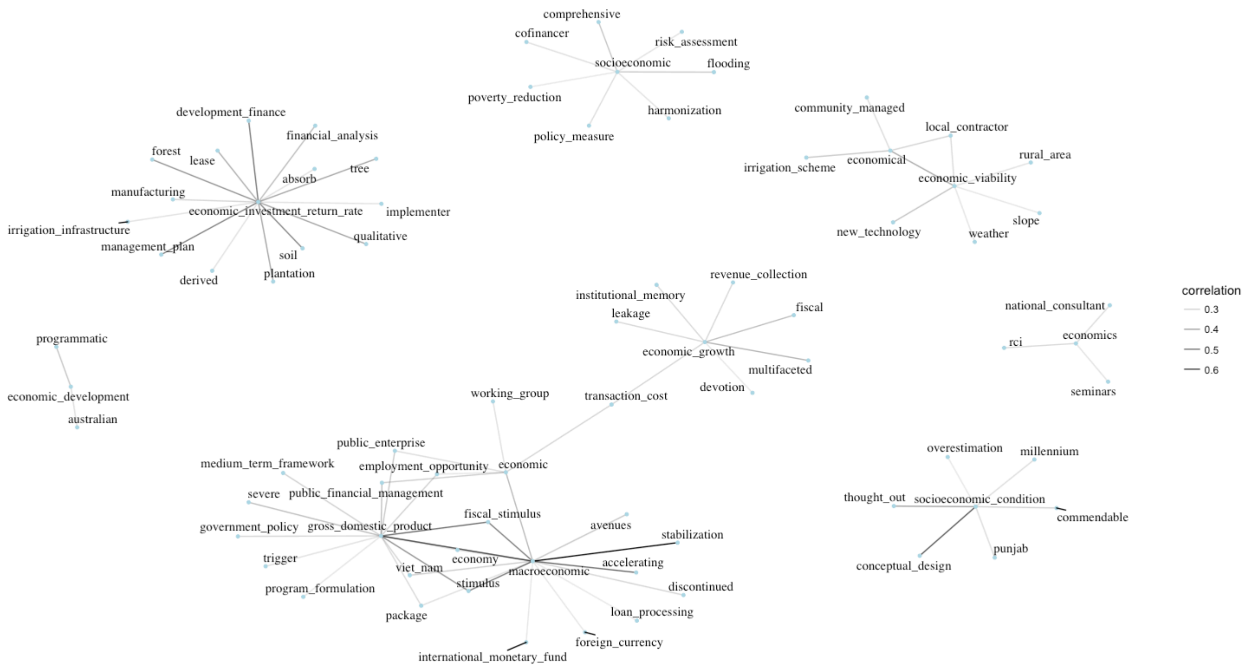

Figure 3: Term correlation for gross domestic product and terms containing 'economic'

Internal evaluations emphasize the influence of policy and institutional environment on project or program outcomes quite clearly. All terms associated with politics are mentioned disproportionately in evaluations of unsuccessful interventions – 'political environment' (log odds: 1.69), 'political interference' (log odds: 1.69), 'political risk' (log odds: 1.69), 'political economy' (log odds: 1.18), 'political commitment' (log odds: 1.00), 'political support' (log odds: 0.71), and 'political will' (log odds: 0.22). This is further reflected in the disproportionate mention of terms such as 'institutional analysis' (log odds: 1.91), 'institutional capability' (log odds: 1.00), 'institutional arrangement' (log odds: 0.83), 'institutional capacity' (log odds: 0.30), and 'institutional memory' (log odds: 0.15) in evaluations of unsuccessful interventions, along with the slightly higher use of 'institutional support' (log odds: -0.10) in evaluations of successful

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interventions. In addition, legal terms such as 'judicial' (log odds: 1.69), 'legal [and] regulatory framework' (log odds: 0.71), 'law' (log odds: 0.57), and 'legal system' (log odds: 0.19) are highlighted more often in evaluations of unsuccessful interventions as well.

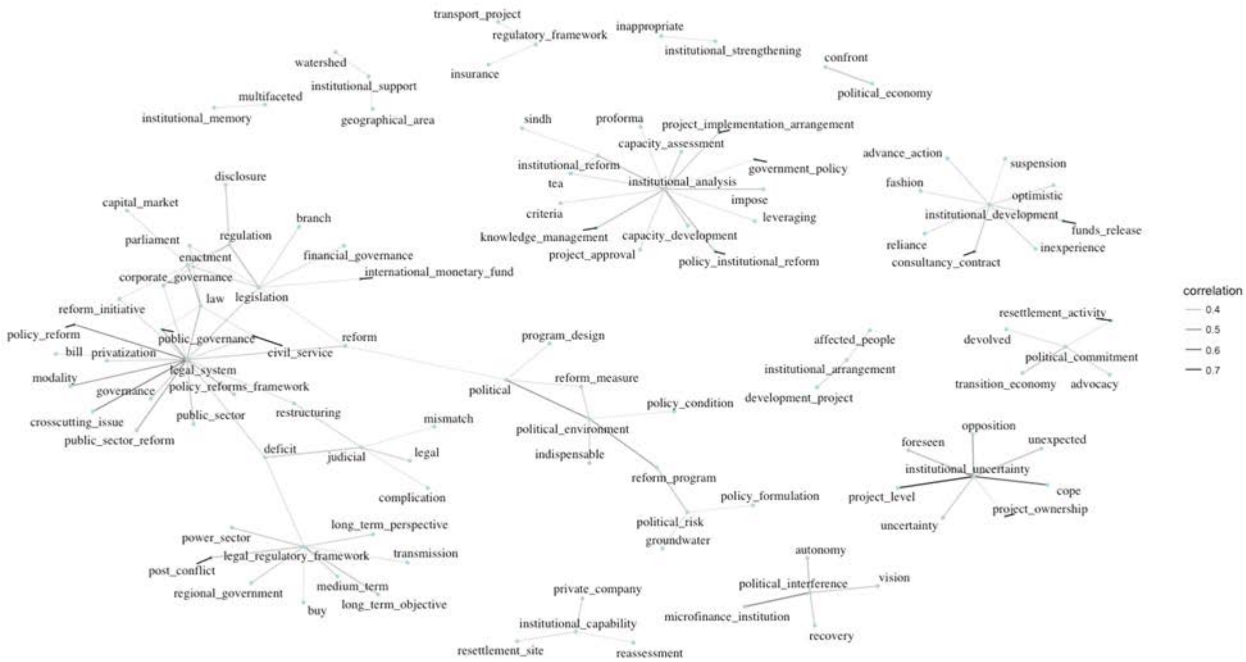

Figure 4: Term correlation for terms related to institution, law, politics, and regulation

Internal evaluations made virtually no mention of 'democracy' or 'authoritarianism'. Neither did they discuss issues pertaining to 'civil liberties' or 'political freedom'. They did, however, mention 'accountability' (log odds: 0.33), more frequently in case of unsuccessful interventions. Accountability was often used in evaluations that also mentioned 'civil service' (correlation: 0.33), 'community empowerment' (correlation: 0.28), 'corruption' (correlation: 0.31), 'public governance' (correlation: 0.31), and 'public sector' (correlation: 0.29). Meanwhile, the term that preceded the use of 'accountability' most often was 'transparency' and the term that

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Tel: (65) 6516 6134 Fax: (65) 6778 1020
Website: www.lkyspp.nus.edu.sg

followed it most often was ‘mechanism’. Thus, even when ‘accountability’ was used in evaluations it was not used in the context of political accountability as H4b might bae implied. Consequently, internal evaluations do not offer sufficient evidence in support of H4a or H4b.

Lessons from completed interventions highlight various project or program characteristics that influence success. First, the disproportionate use of terms such as ‘financially viable’ (log odds: 1.40), ‘overambitious’ (log odds: 1.40), ‘program objective’ (log odds: 1.22), ‘optimistic’ (log odds: 1.14), ‘economic viability’ (log odds: 1.00), ‘unclear’ (log odds: 0.77), and ‘complexity’ (log odds: 0.74) in evaluations of unsuccessful interventions not only supports the view that complexity is associated with lower likelihood of success but also suggests that, more broadly, project or program selection and objectives setting is important for aid effectiveness. In addition, ‘change management’ (log odds: -1.14) and ‘risk management’ (log odds: 0.89) also play a key role, with ‘scope change[s]’ (log odds: 1.00), ‘ad hoc’ (log odds: 0.77), ‘risk’ (log odds: 0.75), and ‘frequent change[s]’ (log odds: 0.21) all appearing more in evaluations of unsuccessful interventions.

Second, while it does appear that evaluations of unsuccessful interventions use the terms ‘discontinued’ (log odds: 1.51) and ‘cancellation’ (log odds: 1.03) more often, issues with ‘commitment’ (log odds: -0.01) have been reported relatively fewer times in evaluations of unsuccessful interventions in comparison to issues with planning and execution, such as ‘conceptual design’ (log odds: 1.91), ‘project approval’ (log odds: 1.77), ‘program formulation’ (log odds: 1.69), ‘loan approval’ (log odds: 1.44), ‘design process’ (log odds: 1.40), ‘lengthy process’ (1.29), ‘funds release’ (log odds: 1.22), and ‘processing time’ (log odds: 1.22). The importance of better planning and procedures is corroborated by the more frequent mention of terms such as 'quality standard[s]' (log odds: -1.64), 'preliminary design' (log odds: -1.49), 'procedure[s] guideline[s]' (log odds: -1.31), 'selection process' (log odds: -1.20), 'detailed design' (log odds: -1.18), and 'project approach' (log odds: -1.08) in evaluations of successful interventions.

Third, while 'capacity constraint[s]' (log odds: 0.80) are indeed associated more with unsuccessful interventions, evaluations identify various types of capacities other than 'management capacity' (log odds: -0.18) that may be responsible for these: 'capacity of the executing or implementing agencies' (log odds: 1.56), 'government capacity' (log odds: 1.56), 'implementation capacity' (log odds: 1.12), 'absorptive capacity' (log odds: 1.00), 'local capacity' (log odds: 1.00), 'financial capacity' (log odds: 0.53), 'staff capacity' (log odds: 0.49), 'institutional capacity' (log odds: 0.30), and 'technical capacity' (log odds: 0.30).

Finally, internal evaluations provide evidence in support of H6 as well. Interestingly, while 'team leader' (log odds: -0.79), 'lead' (log odds: -0.38), 'leadership' (log odds: -0.24), and 'leader' (log odds: -0.18) are used more in evaluations of successful interventions, 'project management' (log odds: 1.00), 'project manager' (log odds: 0.93), 'manager' (log odds: 0.62), and 'project management unit' (log odds: 0.29). Moreover, they highlight the importance of the team work with the higher use of terms such as 'collaboration' (log odds: -1.10), 'working relationship' (log odds: -0.95), and 'cooperation' (log odds: -0.57) in evaluations of successful interventions. Further, the entire 'project team' (log odds: -0.26), including borrowing or recipient countries, executing or implementing agencies, (domestic and international) consultants, contributes to project or program success. Lessons from internal evaluations suggest that other donor agencies may also play a role.

469C Bukit Timah Road
Oei Tiong Ham Building
Singapore 259772
Tel: (65) 6516 6134 Fax: (65) 6778 1020
Website: www.lkyspp.nus.edu.sg

For example, while the ADB has worked well with the 'World Bank' (log odds: -1.81), its record with 'AusAID' has been less successful (log odds: 0.71).

5. Discussion and conclusion

The purpose of this study was to compare lessons from external evaluations of project and program success with those from internal evaluations to understand the need and implications of various types of evaluations for policy learning. To do this, it set forth micro, meso, and macro level hypotheses on project and program success in the case of foreign aid based on a literature review and tested them by using text mining on lessons learnt from internal evaluations of the ADB in over 950 interventions in 38 countries during 1995-2016. The study found evidence within lessons of internal evaluations to corroborate the association of project or program success with specific countries and sectors; supporting policy and institutional environment; and, better project management and supervision. It also found evidence to support the posited relationship between project or program success and characteristics such as selection and scope, early termination, and capacity constraints.

This study also provided evidence beyond these hypotheses to enhance understanding of aid effectiveness that can contribute to learning in the ADB and other agencies to increase the likelihood of project or program success. First, it highlighted potential sources of subnational and subsectoral variation in project or program outcomes. Second, it noted key issues in project planning and execution that receive disproportionate mention in evaluations of unsuccessful interventions. Third, it identified various capacities that can prevent project or program failure

469C Bukit Timah Road
Oei Tiong Ham Building
Singapore 259772
Tel: (65) 6516 6134 Fax: (65) 6778 1020
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(and their relative log odds). Fourth, it emphasized the role not just of the team leader but the entire project team, including other donor agencies that are involved in the project or program.

However, the study did not find sufficient evidence to support the previously observed macro level correlations between project or program success and per capita GDP or its growth rate, civil liberties and political freedom, and political accountability. Thus, while internal evaluations are apt for identifying and highlighting micro level variables such as specific processes or procedures that cause delay, interactions within the project team, or day-to-day influence of politics on project performance, they are limiting for abstracting such issues to draw inferences regarding (macro level) theoretical constructs that may influence aid effectiveness. Instead, they complement external evaluations and efforts at macro level theory building by detecting omitted variables, clarifying causal relationships, and testing causal mechanisms. For example, while external evaluations have only hinted at the relationship between project and program success and capacities, internal evaluations provide data on different types of capacities that are relevant for different types of projects or programs.

Learning from past projects, programs, or policies, then, requires a more nuanced approach than has been suggested in existing literature. A multi-level evaluation programme, consisting of macro, meso, and micro evaluations that complement one another, would allow each layer to pick up elements of success and failure that the others do not. If appropriately used in future projects, such lessons can contribute to greater project and program success in the future.

Lee Kuan Yew School of Public Policy
National University of Singapore

469C Bukit Timah Road
Oei Tiong Ham Building
Singapore 259772
Tel: (65) 6516 6134 Fax: (65) 6778 1020
Website: www.lkyspp.nus.edu.sg

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Oei Tiong Ham Building
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